

LABORATORY SESSION 1: PARTICLE CONNECTED TO A SPOOL

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Abstract

This scientific report is based on an thorough and detailed numeric and physical analysis of different cases of the "Laboratory session 1: Particle Connected to a Spool". This problem consists on a particle P connected with a inextensible massless string to a cylindrical spool of radius a that rotates at a constant angular speed ω . Following the provided laboratory guide, the problem must be parametrized by angle $\phi(t)$, implying just one degree of freedom. As this problem is not analytically integrable in general, we will need to use MATLAB in order to get the position and velocity of P as a function of time in some different cases, while providing the necessary plots and explanations, including, mechanical energy, tension, trajectory and the states variables that take part in these situations.

1 Introduction

As has already been summarized. This laboratory session investigates the dynamics of a particle P of mass m attached to an inextensible massless string of total length l , rolled up around a rotating cylinder of radius a .

The spool system gives us a moving tangency point $B(t)$ and a unspooled string length $\xi(t)$ that, as seen, depends on time. These characteristics will give us different physical phenomena, including non-conservative energy exchange when the spool rotates ($\omega \neq 0$), velocity singularities during string coiling ($\xi \rightarrow 0$), and tether slackening ($T \leq 0$).

The report is structured in 11 different items:

- **Item 1:** Verification of Configuration Degrees of Freedom (C.D.O.F.) and kinematic constraint.
- **Item 2:** Kinematic derivation of position, velocity, and acceleration vectors of particle P .
- **Item 3:** Vectorial representation of applied forces (tension and gravity).
- **Item 4:** Derivation of the differential equation of motion for $\phi(t)$ and string tension $T(t)$.
- **Items 5 & 6:** State-space formulation and event detection (`stopfun`) setup for MATLAB.
- **Items 7 & 8:** Numerical integration and trajectory visualization for all 5 study cases.
- **Items 9–11:** Mechanical energy analysis, physical discussion of critical events, and modeling of post-slackening ballistic motion.

2 Analytical Derivation and Methodology

2.1 Item 1: Degrees of Freedom and Kinematic Constraint

We consider a cylindrical spool of radius a centered at an origin O in an inertial cartesian reference frame $(O; x, y)$. The spool rotates counterclockwise at constant angular velocity $\vec{\omega} = \omega \hat{k}$. The point of tangency between the unspooled string segment and the cylinder surface is B . The radius OB forms an angle $\phi(t)$ with the horizontal axis Ox .

Let l be the total length of the inextensible string. The unspooled string length $\xi(t)$ decreases as string wraps around the spool. The kinematic constraint equation relating $\xi(t)$ and $\phi(t)$ is:

$$\xi(t) = \xi_0 - a(\omega t - \phi(t)) \quad (1)$$

where $\xi_0 = \xi(0)$ is the unspooled length at $t = 0$ when $\phi(0) = 0$.

Verification of Configuration Degrees of Freedom (C.D.O.F.): The configuration of particle P is initially described by 2 generalized coordinates (ξ, ϕ) . However, these coordinates are constrained by the holonomic equation Eq. (1). Therefore:

$$\text{C.D.O.F.} = 2 - 1 = 1 \quad (2)$$

The system possesses **1 configuration degree of freedom**, parametrized uniquely by angle $\phi(t)$. Furthermore, because the constraint equation Eq. (1) contains explicit time t through the spool rotation term ωt , the system is rheonomic (time-dependent).

2.2 Item 2: Kinematic Vectors of Particle P

Differentiating Eq. (1) with respect to time t yields the rate of change of unspooled string length:

$$\dot{\xi}(t) = a\dot{\phi}(t) - a\omega \quad (3)$$

To derive the kinematic vectors, we define a local polar unit basis attached to the rotating radius OB :

$$\hat{e}_r = \cos \phi \hat{i} + \sin \phi \hat{j} \quad (4)$$

$$\hat{e}_\phi = -\sin \phi \hat{i} + \cos \phi \hat{j} \quad (5)$$

Their time derivatives are given by:

$$\dot{\hat{e}}_r = \dot{\phi} \hat{e}_\phi, \quad \dot{\hat{e}}_\phi = -\dot{\phi} \hat{e}_r \quad (6)$$

Inverting Eqs. (4)–(5) gives the Cartesian unit vectors in terms of the polar basis:

$$\hat{i} = \hat{e}_r \cos \phi - \hat{e}_\phi \sin \phi, \quad \hat{j} = \hat{e}_r \sin \phi + \hat{e}_\phi \cos \phi \quad (7)$$

2.2.1 Position Vector $\vec{r}_P$

The position of tangency point B is $\vec{r}_B = a\hat{e}_r$. The straight string segment BP of length ξ extends perpendicularly along $-\hat{e}_\phi$. Hence:

$$\vec{r}_P(t) = a\hat{e}_r - \xi\hat{e}_\phi \quad (8)$$

In Cartesian coordinates, substituting Eq. (7):

$$x_P(t) = a \cos \phi + \xi \sin \phi, \quad y_P(t) = a \sin \phi - \xi \cos \phi \quad (9)$$

2.2.2 Velocity Vector $\vec{v}_P$

Differentiating $\vec{r}_P$ with respect to time and applying Eq. (6):

$$\vec{v}_P = \frac{d\vec{r}_P}{dt} = a\dot{\hat{e}}_r - \dot{\xi}\hat{e}_\phi - \xi\dot{\hat{e}}_\phi = a\dot{\phi}\hat{e}_\phi - \dot{\xi}\hat{e}_\phi + \xi\dot{\phi}\hat{e}_r \quad (10)$$

Substituting $\dot{\xi} = a\dot{\phi} - a\omega$ from Eq. (3):

$$\vec{v}_P = a\dot{\phi}\hat{e}_\phi - (a\dot{\phi} - a\omega)\hat{e}_\phi + \xi\dot{\phi}\hat{e}_r \quad (11)$$

Canceling the $a\dot{\phi}\hat{e}_\phi$ terms yields the final velocity vector:

$$\vec{v}_P(t) = \xi\dot{\phi}\hat{e}_r + a\omega\hat{e}_\phi \quad (12)$$

2.2.3 Acceleration Vector $\vec{a}_P$

Differentiating $\vec{v}_P$ with respect to time:

$$\vec{a}_P = \frac{d\vec{v}_P}{dt} = \frac{d}{dt} \left(\xi \dot{\phi} \right) \hat{e}_r + \xi \dot{\phi} \dot{\hat{e}}_r + a\omega \dot{\hat{e}}_\phi = (\dot{\xi} \dot{\phi} + \xi \ddot{\phi}) \hat{e}_r + \xi \dot{\phi}^2 \hat{e}_\phi - a\omega \dot{\phi} \hat{e}_r \quad (13)$$

Substituting $\dot{\xi} = a\dot{\phi} - a\omega$ into the $\hat{e}_r$ component:

$$\vec{a}_P = \left[(a\dot{\phi} - a\omega) \dot{\phi} + \xi \ddot{\phi} - a\omega \dot{\phi} \right] \hat{e}_r + \xi \dot{\phi}^2 \hat{e}_\phi \quad (14)$$

Grouping like terms yields the acceleration vector:

$$\vec{a}_P(t) = \left(a\dot{\phi}^2 + \xi \ddot{\phi} - 2a\omega \dot{\phi} \right) \hat{e}_r + \xi \dot{\phi}^2 \hat{e}_\phi \quad (15)$$

2.3 Item 3: Mathematical Expression of Forces

The forces acting on particle P are:

1. **String Tension Force $\vec{T}$:** Directed along string segment BP toward tangency point B ($+\hat{e}_\phi$):

$$\vec{T} = T \hat{e}_\phi \quad (16)$$

2. **Gravitational Force $\vec{F}_g$:** Acting vertically downward ($-\hat{j}$). Expressed in the local polar basis using Eq. (7):

$$\vec{F}_g = -mg \hat{j} = -mg(\sin \phi \hat{e}_r + \cos \phi \hat{e}_\phi) \quad (17)$$

2.4 Item 4: Equations of Motion and String Tension

Applying Newton's Second Law ($\sum \vec{F} = m\vec{a}_P$):

$$-mg \sin \phi \hat{e}_r + (T - mg \cos \phi) \hat{e}_\phi = m \left(a\dot{\phi}^2 + \xi \ddot{\phi} - 2a\omega \dot{\phi} \right) \hat{e}_r + m\xi \dot{\phi}^2 \hat{e}_\phi \quad (18)$$

1. Radial Component ($\hat{e}_r$):

$$m \left(a\dot{\phi}^2 + \xi \ddot{\phi} - 2a\omega \dot{\phi} \right) = -mg \sin \phi \quad (19)$$

Dividing by m and solving for angular acceleration $\ddot{\phi}$ gives us the explicit second-order non-linear differential equation, independent of tension T :

$$\ddot{\phi} = \frac{-g \sin \phi - a\dot{\phi}^2 + 2a\omega \dot{\phi}}{\xi} \quad (20)$$

2. Tangential Component ($\hat{e}_\phi$):

$$T - mg \cos \phi = m\xi \dot{\phi}^2 \quad (21)$$

Solving for string tension $T(t)$:

$$T = m\xi \dot{\phi}^2 + mg \cos \phi \quad (22)$$

2.5 Items 5 & 6: State-Space System and MATLAB Setup

To solve Eq. (20) numerically with MATLAB's `ode45`, the second-order ODE is recast into a first-order state-space system with state vector $\mathbf{X} = [X_1; X_2] = [\phi; \dot{\phi}]$:

$$\frac{d\mathbf{X}}{dt} = \begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \end{bmatrix} = \begin{bmatrix} X_2 \\ \frac{-g \sin(X_1) - aX_2^2 + 2a\omega X_2}{\xi_0 - a(\omega t - X_1)} \end{bmatrix} = \mathbf{f}(t, \mathbf{X}) \quad (23)$$

Integration must stop if the particle hits the spool ($\xi \leq 0$) or if the string loses tension ($T \leq 0$). An event function `stopfun` is passed to `ode45`:

$$\text{value} = \begin{bmatrix} \xi(t) \\ T(t) \end{bmatrix}, \quad \text{isterminal} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \text{direction} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \quad (24)$$

2.6 Item 9: Total Mechanical Energy Formulation

The total mechanical energy $E(t) = E_k + E_p$ is calculated as:

$$E(t) = \frac{1}{2}m\|\vec{v}_P\|^2 + mgy_P = \frac{1}{2}m\left(\xi^2\dot{\phi}^2 + a^2\omega^2\right) + mg(a\sin\phi - \xi\cos\phi) \quad (25)$$

3 Study Cases and Results

Numerical simulations were executed using parameters $g = 9.81 \text{ m/s}^2$, $m = 0.1 \text{ kg}$, $a = 0.2 \text{ m}$, $\phi(0) = 0$, and $\xi(0) = 1 \text{ m}$. Table 1 summarizes all 5 study cases.

Table 1: Parameters, initial conditions, and numerical outcomes for the 5 study cases.

Case	ω (rad/s)	v_0 (m/s)	$\dot{\phi}(0)$ (rad/s)	Initial Energy E_0 (J)	Termination Event
Case 1	0.1	0.0	0.0	-0.9790	Forced winding / Spool impact ($\xi = 0$)
Case 2	0.0	-1.0	-1.0	-0.9310	Stable pendulum oscillation
Case 3	0.0	-10.0	-10.0	4.0210	Spool impact ($\xi = 0$) at $t \approx 0.38 \text{ s}$
Case 4	0.0	4.0	4.0	-0.1810	Loss of tension ($T = 0$) at $t \approx 0.62 \text{ s}$
Case 5	0.1	1.0	1.0	-0.9290	Forced winding / Spool impact ($\xi = 0$)

3.1 Individual Case Overview

- **Case 1** ($\omega = 0.1 \text{ rad/s}$, $v_0 = 0$): Particle P starts from rest relative to tangency point B . Spool rotation forces string winding, leading to eventual impact ($\xi = 0$).
- **Case 2** ($\omega = 0$, $v_0 = -1 \text{ m/s}$): With a stationary spool, the unspooled length remains constant ($\xi = 1 \text{ m}$), producing stable simple pendulum oscillations.
- **Case 3** ($\omega = 0$, $v_0 = -10 \text{ m/s}$): High launch velocity causes rapid string wrapping around the cylinder. Integration terminates upon spool collision ($\xi = 0$).
- **Case 4** ($\omega = 0$, $v_0 = 4 \text{ m/s}$): Upward launch causes P to enter the upper quadrant where gravity induces string slackening ($T = 0$).
- **Case 5** ($\omega = 0.1 \text{ rad/s}$, $v_0 = 1 \text{ m/s}$): Combines an initial launch velocity with active spool rotation, resulting in forced wrapping and oscillating string tension until impact ($\xi = 0$).

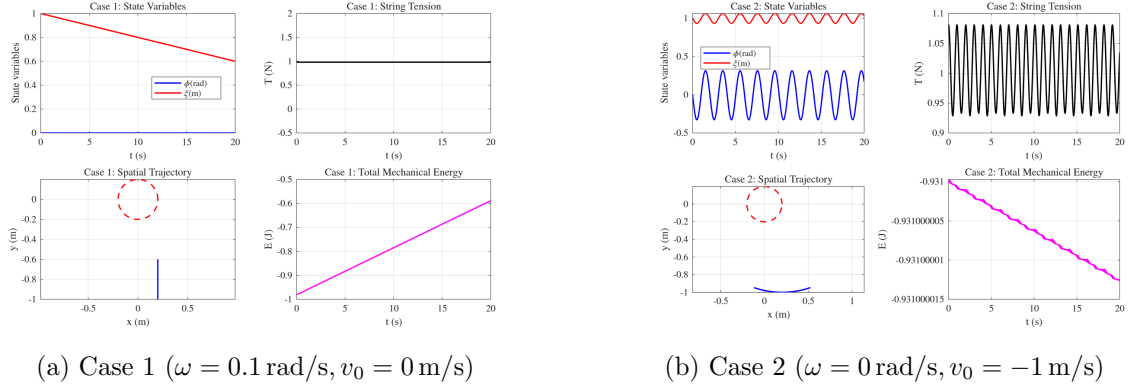


Figure 1: Numerical integration results for Cases 1 and 2.

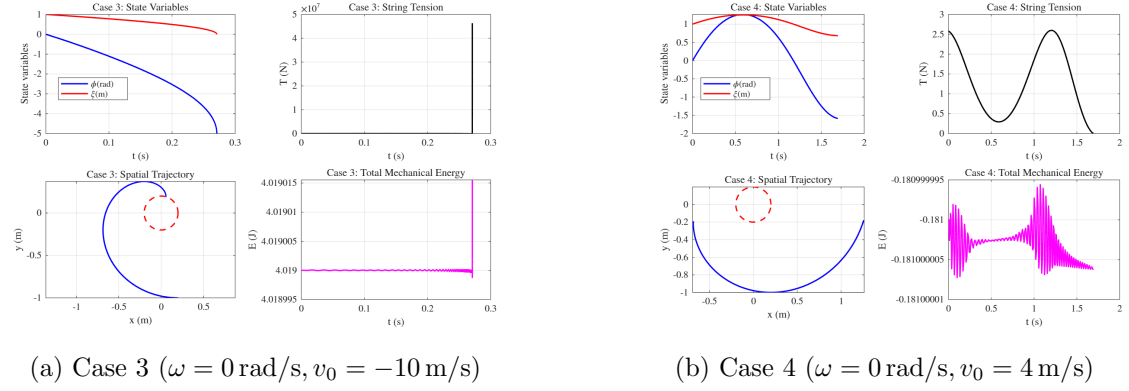
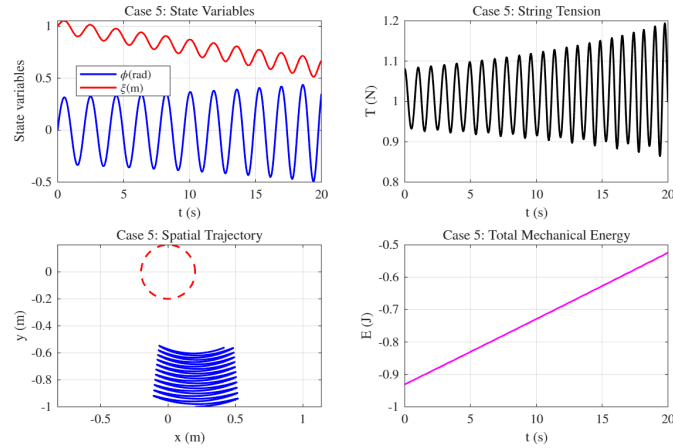


Figure 2: Numerical results for wrapping impact (Case 3) and string slackening (Case 4).


 Figure 3: Numerical integration results for Case 5 ($\omega = 0.1 \text{ rad/s}$, $v_0 = 1 \text{ m/s}$), depicting state variables, string tension, Oxy trajectory, and energy evolution.

4 Physical Discussion

4.1 Item 9: Proof of Mechanical Energy Conservation

The rate of work done by non-conservative forces determines energy conservation. The mechanical power input by string tension $\vec{T} = T\hat{e}_\phi$ is:

$$P_T = \vec{T} \cdot \vec{v}_P = (T\hat{e}_\phi) \cdot (\xi\dot{\phi}\hat{e}_r + a\omega\hat{e}_\phi) = Ta\omega \quad (26)$$

- **Cases with $\omega = 0$ (Cases 2, 3, 4):** When the spool is stationary, $P_T = 0$ because tension $\vec{T}$ is strictly perpendicular to velocity $\vec{v}_P = \xi\dot{\phi}\hat{e}_r$. The tether performs zero work ($\delta W = 0$), and total mechanical energy $E(t)$ is **strictly conserved** ($dE/dt = 0$).
- **Cases with $\omega \neq 0$ (Cases 1, 5):** When the spool rotates, $P_T = Ta\omega \neq 0$. The rotating spool performs work on the mass, acting as an external energy source/sink and causing $E(t)$ to fluctuate over time.

4.2 Item 10: Physical Mechanisms of Critical Events

4.2.1 Angular Velocity Divergence at Spool Impact (Cases 3 and 5)

In Cases 3 and 5, the particle wraps around the cylinder until $\xi \rightarrow 0$. From Eq. (20), $\ddot{\phi} = (-g \sin \phi - a\dot{\phi}^2 + 2a\omega\dot{\phi})/\xi$. As $\xi \rightarrow 0$, the vanishing denominator causes angular acceleration $\ddot{\phi}$ and angular velocity $\dot{\phi}$ to diverge. Physically, energy conservation for $\omega = 0$ requires $E = \frac{1}{2}m\xi^2\dot{\phi}^2 + mgy_P = E_0$. Because potential energy remains bounded ($|mgy_P| \leq mga$), the product $\xi\dot{\phi}$ must remain finite. Consequently:

$$|\dot{\phi}(t)| \sim \frac{\sqrt{2(E_0 - mgy_P)/m}}{\xi(t)} \xrightarrow{\xi \rightarrow 0} \infty \quad (27)$$

The angular velocity diverges as $\mathcal{O}(1/\xi)$, causing the mass to whip violently around the spool immediately prior to collision.

4.2.2 String Slackening (Case 4)

In Case 4 ($v_0 = 4 \text{ m/s}$), gravity decelerates the particle near the top of its swing ($\cos \phi < 0$). At $t \approx 0.62 \text{ s}$, the centrifugal term $m\xi\dot{\phi}^2$ can no longer overcome the inward gravitational pull, causing string tension $T = m\xi\dot{\phi}^2 + mg \cos \phi$ to drop to zero.

4.3 Item 11: Open Discussion – Post-Slackening Ballistic Modeling

When string tension reaches zero ($T = 0$), the flexible tether cannot support compression and goes slack. Continuing numerical integration past t_{slack} requires switching from constrained 1-DoF dynamics to a **2-DoF unconstrained ballistic free-fall model**:

4.3.1 Phase 1: Free Ballistic Flight ($t > t_{\text{slack}}$)

During slackness, the particle moves under gravity alone ($\vec{F} = -mg\hat{j}$):

$$\ddot{x}_P(t) = 0, \quad \ddot{y}_P(t) = -g \quad (28)$$

Initial conditions for this parabolic phase are inherited directly from the state at t_{slack} :

$$\vec{r}_P(t_{\text{slack}}) = a \cos \phi_s \hat{i} + a \sin \phi_s \hat{j} - \xi_s \hat{e}_{\phi,s}, \quad \vec{v}_P(t_{\text{slack}}) = \xi_s \dot{\phi}_s \hat{e}_{r,s} + a\omega \hat{e}_{\phi,s} \quad (29)$$

4.3.2 Phase 2: Re-Tensioning and Inelastic Impulse

As point $B(t)$ continues rotating on the spool, the distance between $P(t)$ and $B(t)$ is monitored:

$$d_{PB}(t) = \|\vec{r}_P(t) - \vec{r}_B(t)\| \quad (30)$$

The string becomes taut again when $d_{PB}(t) = \xi(t)$. At this instant $t_{re-tension}$, an impulsive jerk occurs. The velocity component parallel to the string is eliminated via an inelastic collision model, and 1-DoF constrained integration resumes using Eq. (20).

5 Conclusions

In this laboratory session, the dynamic behavior of a particle connected to a rotating spool was successfully modeled and analyzed:

1. The system possesses 1 configuration degree of freedom parametrized by $\phi(t)$, with explicit time dependence introduced by spool rotation ω .
2. MATLAB integration using `ode45` coupled with `stopfun` accurately detects spool collision ($\xi = 0$) and tether slackening ($T = 0$).
3. Mechanical energy is conserved if and only if $\omega = 0$. For $\omega \neq 0$, the rotating spool performs non-conservative work via string tension.
4. During wrapping ($\xi \rightarrow 0$), angular velocity diverges as $\mathcal{O}(1/\xi)$ to satisfy energy conservation.
5. String slackening ($T = 0$) requires transitioning to a 2-DoF ballistic free-fall ODE until tether re-tensioning occurs.

References

1. Department of Aerospace Engineering, UC3M, *Mechanics Applied to Aerospace Engineering: Laboratory Guidelines*, Version 2026.
2. I. Fernández Tena, *Lab Session 1: Problem of Particle Connected to a Spool and MATLAB Integration Guidelines*, Course Notes, 2026.